

EDS Practical :05

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Batch: D22

5.1.1 Stacked plot

Code:

```
import matplotlib.pyplot as plt
import pandas as pd

# Data for Months and Temperature for three cities
data = {
    'Month': ['January', 'February', 'March', 'April', 'May',
              'June', 'July', 'August', 'September', 'October', 'November',
              'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18,
                           12, 8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5,
                           3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
```

```
}
```

Write your code...

```
df = pd.DataFrame(data)
```

```
plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_
B_Temperature'],df['City_C_Temperature'])
```

```
plt.xlabel('Month')
```

```
plt.ylabel('Temperature')
```

```
plt.title('TemperatureVariation')
```

```
plt.legend(loc='upper left')
```

```
plt.show()
```

Average time

0.238 s

238.00 ms



Maximum time

0.238 s

238.00 ms



✓ 1 out of 1 shown test case(s) passed

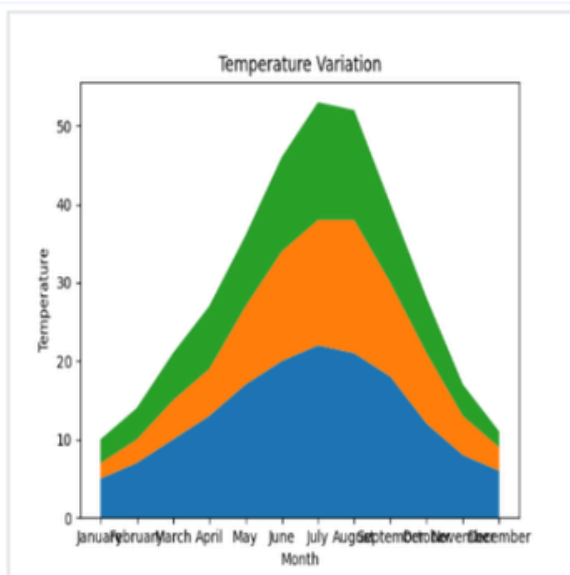
✓ Test case 1

238 ms

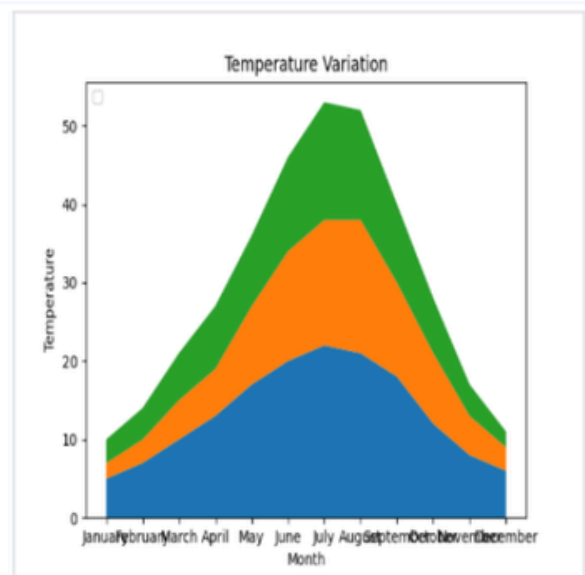
Debug



Expected output



Actual output



5.2.1 Titanic Dataset

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')
# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))
# write the code..
##plot 1:count of passengers by class
axes[0,
0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(), color='skyblue')
axes[0, 0].set_title("Passenger Class Distribution")
axes[0,0].set_xlabel("Pclass")
axes[0,0].set_ylabel("Count")
axes[0, 1].pie(df['Gender'].value_counts(), labels=df['Gender'].value_counts().index, autopct='% 1.1f%%', colors=['lightblue', 'lightcoral'])
axes[0,1].set_title("Gender Distribution")
#plot 3:Age distribution
axes[1, 0].hist(df['Age'].dropna(), bins=8, color='lightgreen')
```

```
edgecolor='black')
```

```
axes[1,0].set_title("Age Distribution")
```

```
axes[1,0].set_xlabel("Age")
```

```
axes[1,0].set_ylabel("Frequency")
```

```
#plot 4:survival count
```

```
axes[1,1].bar(df['Survived'].value_counts().index, df['Survived'].value_counts(), color=['lightblue', 'lightcoral'])
```

```
axes[1,1].set_title("Survival  
Count")
```

```
axes[1,1].set_xlabel("Survived (0 = No, 1 = Yes)")
```

```
axes[1,1].set_ylabel("Count")
```

```
#plot 5:fare vs age
```

```
axes[2,0].scatter(df['Age'], df['Fare'], color='orange', edgecolor='black')
```

```
axes[2,0].set_title("Fare vs Age")
```

```
axes[2,0].set_xlabel("Age")
```

```
axes[2,0].set_ylabel("Fare")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output :

Average time

0.900 s
900.00 ms



Maximum time

0.900 s
900.00 ms



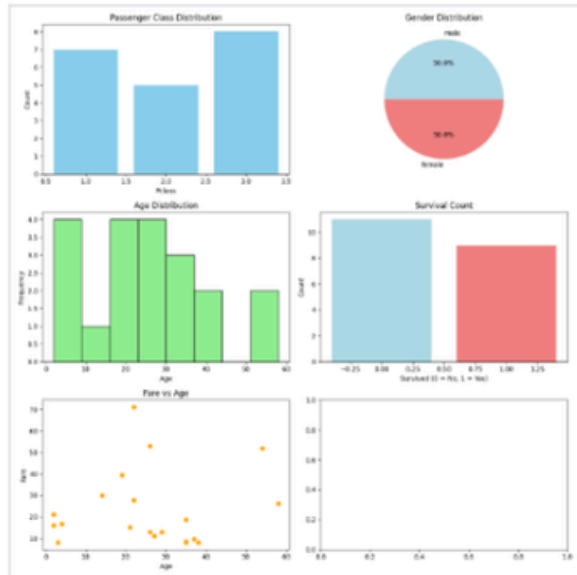
✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 900 ms

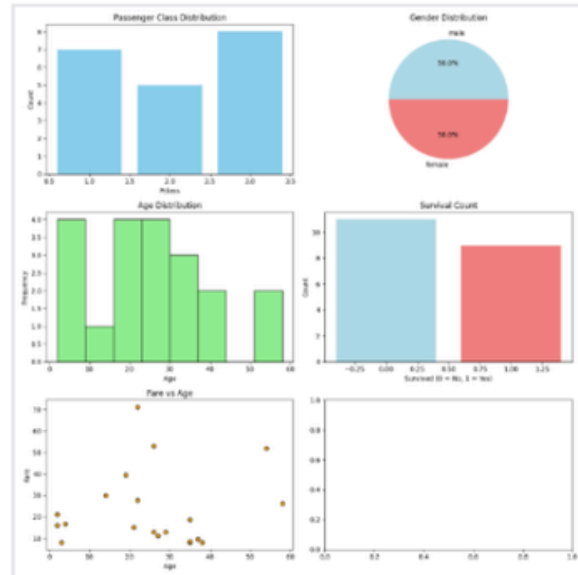
Debug



Expected output



Actual output



5.2.2 Histogram of Passenger information of Titanic

Code:

```
#write your code here for histogram
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning
```

```

data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True)
data.drop('Cabin', axis=1, inplace=True)
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True )
# Write your code here for
# Write your code here for Histogram
#write your code here for histogram
plt.hist(data['Age'],bins=30,edgecolor='k')
#plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('AgeDistribution')
plt.show()
Output :
```

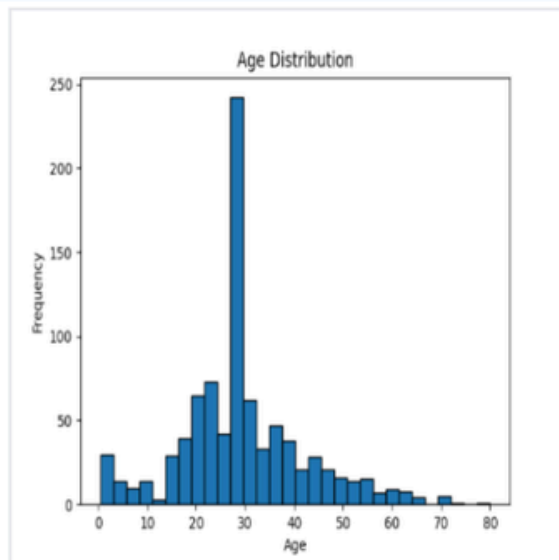
Average time
0.265 s
265.00 ms

Maximum time
0.265 s
265.00 ms

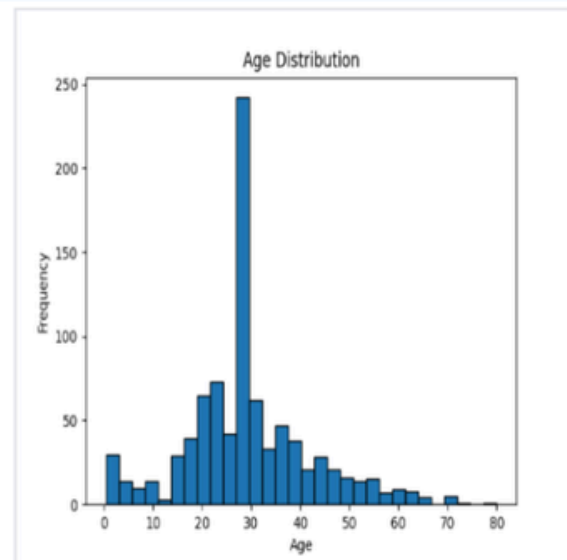
✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 265 ms Debug

Expected output



Actual output



5.2.3 Bar plot of survival rate of passengers

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

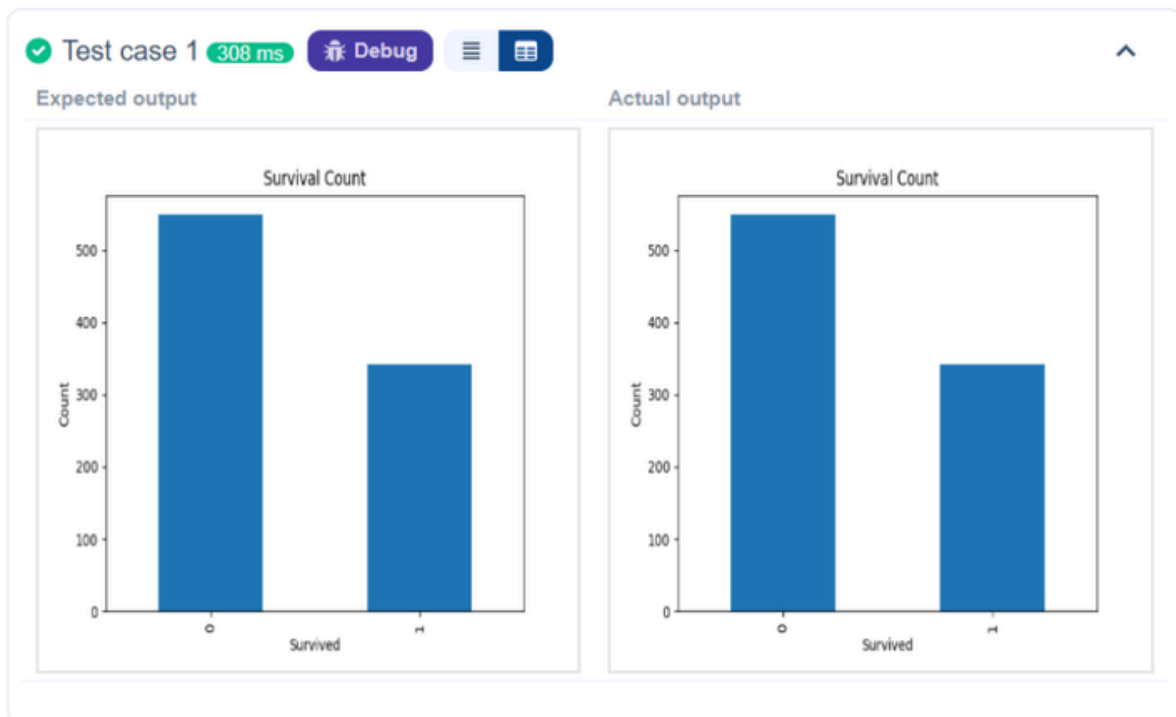
```
# Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(),inplace=True)
```

```
data['Embarked'].fillna(data['Embarked'].mode()[0],
```

```
inplace=True)
data.drop('Cabin', axis=1, inplace=True)
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
dropp_first =True )
# Write your code here for Bar Plot for Survival Rate
survival_counts = data['Survived'].value_counts()
survival_counts.plot(kind='bar')
plt.title('Survival Count')
plt.xlabel('Survived')
plt.ylabel('Count')
plt.show()
Output :
```

Average time **0.308 s** 308.00 ms Maximum time **0.308 s** 308.00 ms **1 out of 1 shown test case(s) passed**



5.2.4 Bar plot for survival by Gender

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
# Data Cleaning
data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True)
```

```
data.drop('Cabin', axis=1, inplace=True)
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)
# Write your code here for Bar Plot for Survival by
Gender
grouped=data.groupby('Sex')
['Survived'].value_counts().unstack()
grouped.plot(kind='bar', stacked=True)
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['NotSurvived','Survived'])
plt.show()
Output :
```

Average time **0.377 s** 377.00 ms Maximum time **0.377 s** 377.00 ms ✓ 1 out of 1 shown test case(s) passed



5.2.5 Bar plot for survival by Pclass

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
# Data Cleaning
data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True)
data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
                        drop_first=True)

# Write your code here for Bar Plot for Survival by Pclass
survival_counts = data.groupby('Pclass')
['Survived'].value_counts().unstack()
survival_counts.plot(kind='bar', stacked=True)
plt.title('Survival by Pclass')
plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output

Average time

0.341 s
341.00 ms



Maximum time

0.341 s
341.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ Test case 1

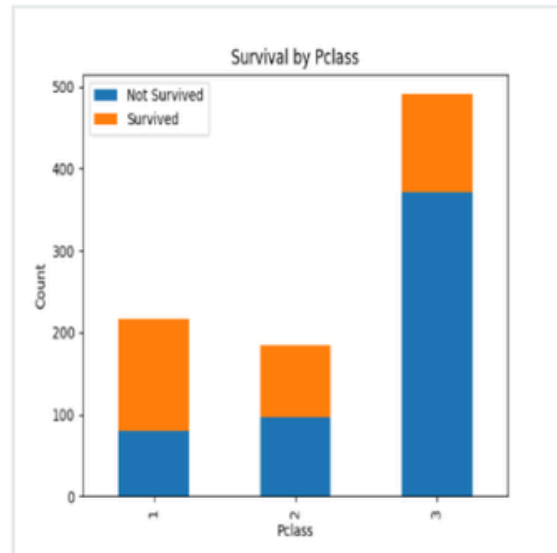
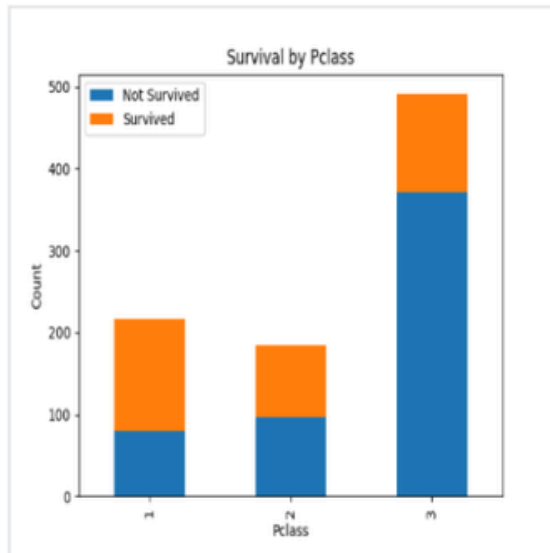
341 ms

Debug



Expected output

Actual output



5.2.6 Bar plot for survival by Embarked

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(),inplace=True)
```

```
data['Embarked'].fillna(data['Embarked'].mode()[0],  
inplace=True)
```

```
data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
                        drop_first=True)

# Write your code here for Bar Plot for Survival by
# Embarked
survival_counts = data.groupby('Embarked_Q')
['Survived'].value_counts().unstack()
survival_counts.plot(kind='bar', stacked=True)
plt.title('Survival by Embarked')
plt.xlabel('Embarked')
plt.ylabel('Count')
plt.legend(['NotSurvived', 'Survived'])
plt.show()
```

Output :



5.2.7 Box plot for Age Distribution

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
# Data Cleaning
data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True)
data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
                        drop_first=True)
# Write your code here for Box Plot for Age by Pclass
data.boxplot(column='Age',by='Pclass')
plt.title('Age by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.suptitle('')
plt.show()
```

Output :

Average time

0.326 s
326.00 ms



Maximum time

0.326 s
326.00 ms



✓ 1 out of 1 shown test case(s) passed

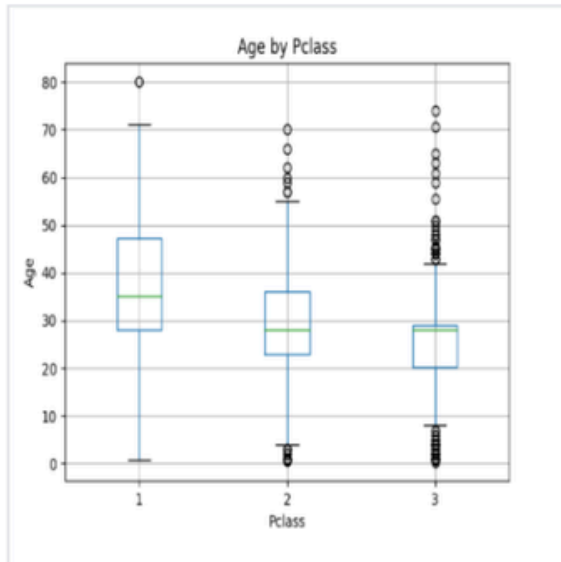
✓ Test case 1

326 ms

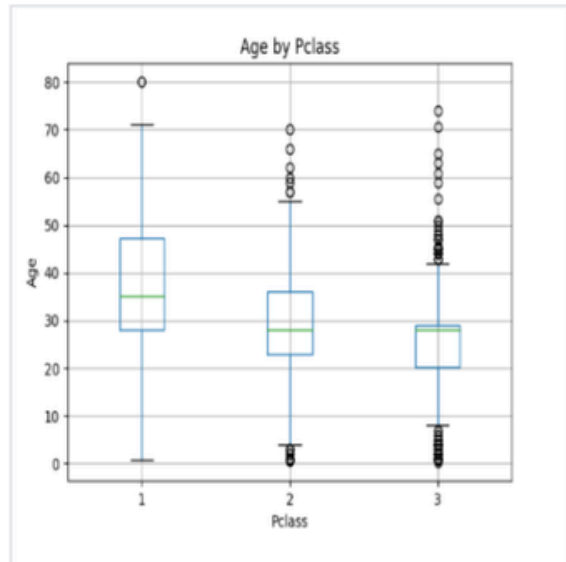
Debug



Expected output



Actual output



5.2.8 Box plot for Age by Survived

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(),inplace=True)
```

```
data['Embarked'].fillna(data['Embarked'].mode()[0],  
inplace=True)
```

```
data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric
```

```
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

```
data = pd.get_dummies(data, columns=['Embarked'],  
drop_first=True)
```

```
# Write your code here for Box Plot for Age by Survived
```

```
data.boxplot(column='Age', by='Survived')
```

```
plt.title('Age by Survival')
```

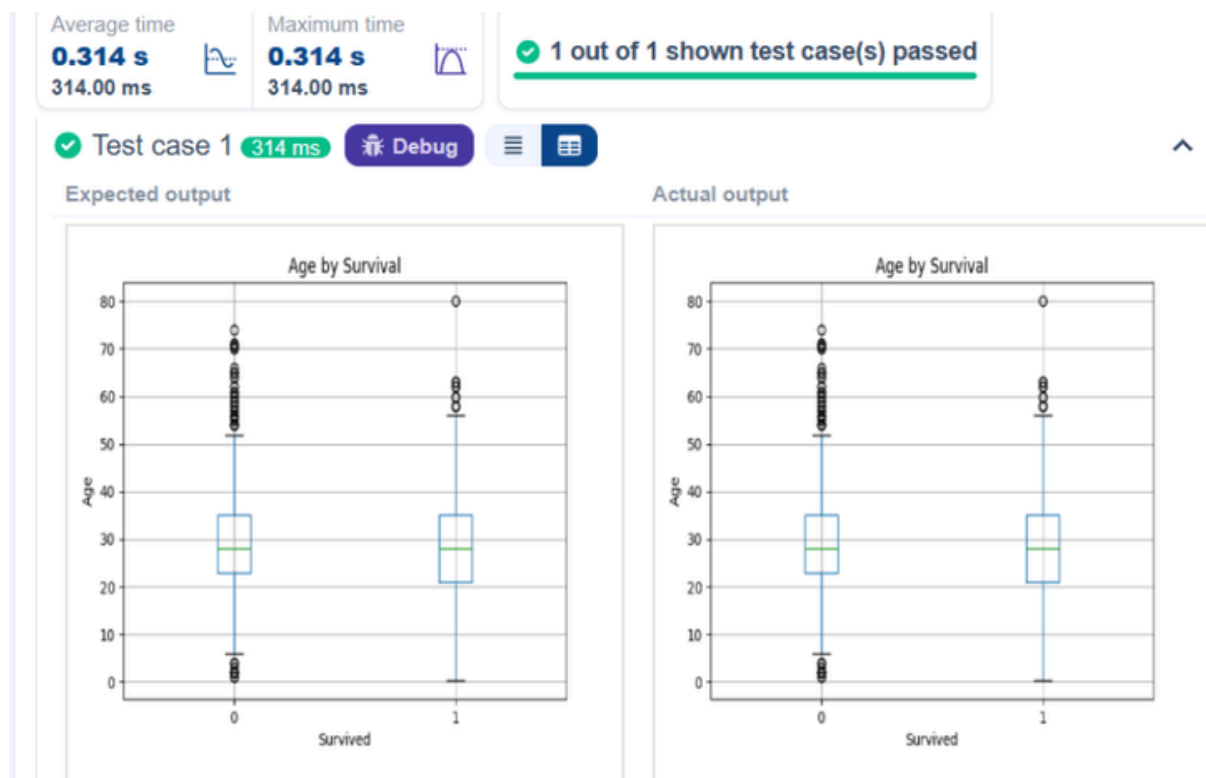
```
plt.xlabel('Survived')
```

```
plt.ylabel('Age')
```

```
plt.suptitle('')
```

```
plt.show()
```

Output :



5.2.9 Box plot for fare by Pclass

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
# Data Cleaning
data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True)
data.drop('Cabin', axis=1, inplace=True)
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)
# Write your code here for Box Plot for Fare by Pclass
data.boxplot(column='Fare', by='Pclass')
plt.title('Fare by Pclass')
plt.suptitle("")
plt.xlabel('Pclass')
plt.ylabel('Fare')
```

plt.show()

Output :



5.2.10 Scatter plot for Age vs Fare

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
# Data Cleaning
data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
```

```
inplace=True)
data.drop('Cabin', axis=1, inplace=True)
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
dropp_first =True )
# Write your code here for Box Plot for Fare by Pclass
plt.scatter(data['Age'],data['Fare'])
plt.title("Age vs. Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
O ut put :
```

Average time

0.320 s
320.00 ms

Maximum time

0.320 s
320.00 ms

✓ 1 out of 1 shown test case(s) passed

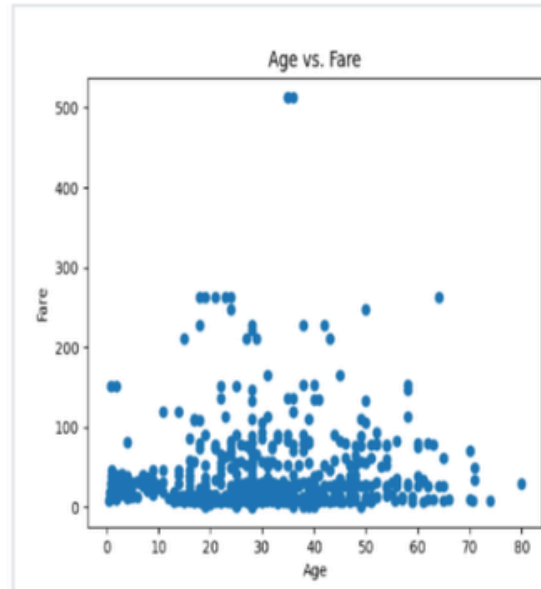
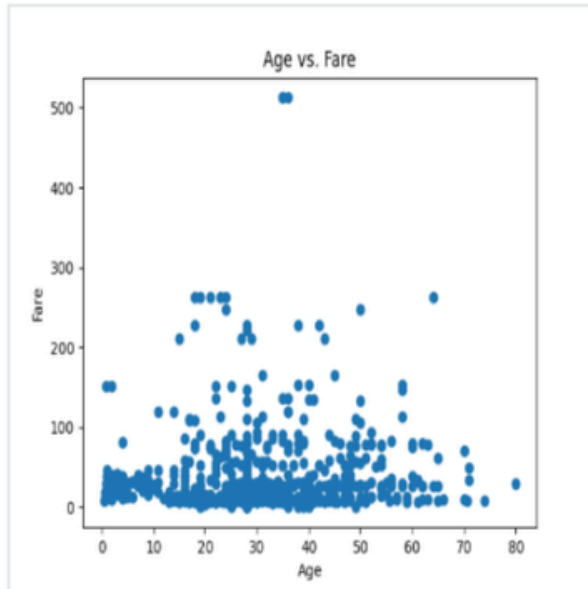
✓ Test case 1 320 ms

Debug



Expected output

Actual output



5.2.11 Scatter plot for Age vs Fare by Survived

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(),inplace=True)
```

```
data['Embarked'].fillna(data['Embarked'].mode()[0],  
inplace=True)
```

```
data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
                        drop_first=True)
# Write your code here for Scatter Plot for Age vs. Fare
# by Survived
colours=data['Survived'].map({0: 'red',1: 'blue'})
plt.scatter(data['Age'], data['Fare'], c=colours)
plt.title("Age vs. Fare by Survival")
plt.xlabel("Age")
plt.ylabel('Fare')
plt.show()
Output :
```

Average time

0.299 s

299.00 ms



Maximum time

0.299 s

299.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 299 ms

Debug



Expected output

Actual output

